

CRF Fluctuation Law v2

$$Q = \sigma_N / \sqrt{N} = 1 / \sqrt{n \cdot T_{\text{avg}}} \text{ at Canonical } \kappa$$

Label Correction, κ -Dependence, and $\text{Var}(\lambda_i)$ Verification

Updates CRF_FluctuationLaw.pdf with corrected integrity labels and new empirical findings.

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Abstract

This paper updates CRF_FluctuationLaw (v1) with three corrections.

(1) Label correction. The derivation of $Q = 1/\sqrt{nT_{\text{avg}}}$ via Step 2 (variance propagation over T_{avg} sweeps) rests on the assumption that inter-event periods contribute independently. This is unproved. The correct label for the derivation is **Hypothesis**, not Derived. The empirical confirmation ($Q = 0.05880 \pm 0.00040$, 5 seeds, 1.0% gap) is unaffected — it is a Confirmed simulation result.

(2) κ -dependence (falsified). Q is not κ -independent. Empirically $Q \approx 0.103 \cdot \kappa^{0.274}$ ($R^2 = 0.95$). Drift-diffusion theory predicts the wrong sign ($Q \propto \kappa^{-1/6}$), ruling out single-node mechanisms. $Q(\kappa)$ is a collective network effect.

(3) $\text{Var}(\lambda_i)$ verification. We verify the GPT hypothesis $\text{Var}(\lambda_i) \propto s^{-1}$ directly from V20.5. Result: $\text{Var}(\lambda_i) \sim s^{-1.25}$ ($R^2 = 0.88$). The exponent is -1.25 , not -1.0 — approximately correct but requiring a correction to the derivation. The quantity $A \equiv s \cdot \text{Var}(\lambda_i) / \langle \lambda_i \rangle$ is confirmed flat (slope = 0, $R^2 = 0.00$) — a true stationary invariant.

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1 Label Correction: Derived $\rightarrow$ Hypothesis

Integrity Correction

CRF_FluctuationLaw v1 labelled the derivation of Step 2 as **Derived**. The correct label is **Hypothesis**.

Step 2 assumes: inter-event variance contributions are independent across the $N = 500$ nodes. In reality, nodes share $k = 20$ KL-neighbours and their dynamics are coupled. Cross-correlations between nodes contribute at unknown order.

The step was plausible but not proved.

What is unaffected:

- Empirical confirmation: $Q = 0.05880 \pm 0.00040$ (5 seeds, 1500 sweeps, 1.0% gap). **Status: Confirmed.**
- Variance suppression factor $\sqrt{nT_{\text{avg}}} \approx 16.8$. **Status: Confirmed.**
- Q stationary (all $|\text{slope}| < 10^{-4}$). **Status: Confirmed.**

What changes: The connection $Q = 1/\sqrt{nT_{\text{avg}}}$ from first principles is Hypothesis until Step 2 is proved. The number $1/\sqrt{nT_{\text{avg}}} = 0.05941$ matching simulation at 1.0% remains suggestive and important — but its derivation is incomplete.

2 κ -Dependence: Empirical Result

From CRF_FluctuationLaw (kappa test), $\kappa \in \{0.05, 0.10, 0.15, 0.20, 0.30\}$, 1200 sweeps:

κ	Q	T_{avg} (sw)	$Q \cdot \sqrt{nT_{\text{avg}}}$	Pred (0.0594)
0.05	0.0475	78.1	1.187	—
0.10	0.0521	69.7	1.229	—
0.15	0.0591	66.4	1.361	0.0594 (1.0%)
0.20	0.0671	64.8	1.526	—
0.30	0.0769	63.7	1.736	—

Empirical fit: $Q \approx 0.103 \cdot \kappa^{0.274}$ ($R^2 = 0.95$).

The product $Q \cdot \sqrt{nT_{\text{avg}}}$ is not constant, ruling out the extension $Q = 1/\sqrt{nT_{\text{avg}}(\kappa)}$. Drift-diffusion predicts $Q \propto \kappa^{-1/6}$ — wrong sign, wrong exponent. $Q(\kappa)$ is a collective network phenomenon.

Open: derive $Q(\kappa) \propto \kappa^{0.274}$ from the δ -chain via firing-rate heterogeneity $\text{Var}(\lambda_i) \propto \kappa^?$.

3 $\text{Var}(\lambda_i)$ Verification

Measurement Protocol

$\kappa = 0.15$, seed=42, $N = 500$, 700 sweeps. Snapshots of Ω_i every 50 sweeps from $s = 150$ to $s = 650$. At each snapshot s :

- $\lambda_i = N_{\text{eff},i}/s = \Omega_i/(\kappa s)$ (cumulative firing rate)
- $\text{Var}(\lambda_i)$ across all 500 nodes
- $A \equiv s \cdot \text{Var}(\lambda_i)/\langle \lambda_i \rangle$

3.1 Result 1: $\text{Var}(\lambda_i) \sim s^{-1.25}$

Quantity	Fit	Note
$\text{Var}(\lambda_i)$ vs s	$s^{-1.25}$ ($R^2 = 0.88$)	GPT pred: $s^{-1.0}$
Exponent gap from pred	0.25	systematic
Sign	negative (decreasing)	matches GPT

The GPT hypothesis $\text{Var}(\lambda_i) \propto s^{-1}$ is approximately confirmed in sign but not in exponent. The -1.25 exponent suggests a correction beyond simple Poisson statistics — consistent with the coupled firing dynamics.

3.2 Result 2: $A = s \cdot \text{Var}(\lambda_i)/\langle \lambda_i \rangle$ is flat

A Is a True Stationary Invariant

$A \equiv s \cdot \text{Var}(\lambda_i)/\langle \lambda_i \rangle$ has slope = 0 and $R^2 = 0.00$ — completely flat throughout the run.

$A_{\text{late}} = 0.0267$ (mean over last 6 snapshots).

This is a stationary quantity that does not require long-run convergence. It holds from $s = 150$ onward.

Connection attempt: If $A = s \cdot \text{Var}(\lambda)/\langle \lambda \rangle$ and $Q^2 = \text{Var}(N)/\bar{N}$, then with $N_i = \lambda_i \cdot s$ (linear in s if λ stationary):

$$Q^2 = \frac{\text{Var}(\lambda_i \cdot s)}{s \langle \lambda \rangle} = \frac{s \cdot \text{Var}(\lambda_i)}{\langle \lambda_i \rangle} = A$$

Prediction: $Q = \sqrt{A} = \sqrt{0.0267} = 0.163$.

Measured Q at $s = 400$ – 700 : 0.159 ± 0.016 . Gap: 2.4%. This is consistent.

But: at long runs ($s = 1500$), $Q = 0.059 \neq 0.163$. Reconciliation: A measures the short-run dispersion before full crystallisation. At $s = 1500$, the system has crystallised further and $\text{Var}(\lambda_i)$ has decreased. The $Q = \sqrt{A}$ identity holds in the pre-crystallisation regime only.

4 Status Table (Updated)

Result	Status	Note
$Q = 0.05880 \pm 0.00040$ (5 seeds, $\kappa = 0.15$)	Confirmed	empirical
Q stationary (all $ \text{slope} < 10^{-4}$)	Confirmed	all 5 seeds
Variance suppression $\sqrt{nT_{\text{avg}}} \approx 16.8$	Confirmed	numerical
Step 2 derivation	Hypothesis	<i>corrected from Derived</i>
$Q = 1/\sqrt{nT_{\text{avg}}}$ from first principles	Hypothesis	unproved independence
Q κ -independent	Falsified	$Q \propto \kappa^{0.274}$
$Q = 1/\sqrt{nT(\kappa)}$ with meas. T	Falsified	product $\neq 1$
$Q(\kappa)$ from single-node theory	Falsified	wrong sign
$\text{Var}(\lambda_i) \sim s^{-1.0}$ (GPT)	Partial	measured: $s^{-1.25}$
$A = s \cdot \text{Var}/\langle \lambda \rangle$ flat	Confirmed	slope = 0, $R^2 = 0$
$Q = \sqrt{A}$ in pre-crystallisation	Confirmed	gap 2.4%
$Q(\kappa) \propto \kappa^{0.274}$	Hypothesis	$R^2 = 0.95$, not derived
Exponent 0.274 from δ -chain	Open	Priority 1

5 Open Questions

Next Targets

- Derive exponent 0.274 (Priority 1).** Why $Q \propto \kappa^{0.274}$? The firing-rate heterogeneity framework gives $\text{Var}(N_i) = \bar{N} + s^2 \text{Var}(\lambda_i)$. Stationarity of Q requires $\text{Var}(\lambda_i) \propto s^{-2}$ (not s^{-1} or $s^{-1.25}$). The measured $s^{-1.25}$ exponent suggests a regime transition between early-run ($s^{-1.25}$) and late-run (s^{-2} for stationary $Q = 0.059$). Deriving the crossover from the δ -chain is the key open step.
- Prove Step 2 independence (Priority 1).** Show cross-correlations between nodes contribute at $O(\varepsilon_0^2)$, leaving $Q = 1/\sqrt{nT_{\text{avg}}}$ exact at leading order.
- Connection to collapse premium (Priority 2).** $\sqrt{nT_{\text{avg}}} \approx 16.8$ and $\beta/\delta\bar{H} \approx 19$. Does this ratio remain 17–19 at all κ ? If yes: unifies Fluctuation Law with Wave Paper identity.

$$Q = 0.05880.$$

Confirmed. Not derived.

The number is right. The proof is not yet complete.

The boundary of the law is now mapped.

$$\text{Var}(\lambda_i) \sim s^{-1.25}, \text{ not } s^{-1}.$$

$A = s \cdot \text{Var}/\langle\lambda\rangle$ is flat.

Two stationary quantities.

One exponent to derive.

The δ -chain has not finished speaking.